

Independent coil and link
endpoint measurements

Combined plant and controller boundary

Plant

$$Li^2/2 + mv^2/2 + kx^2/2$$

$$v_s = F_s = 0$$

$$P_r = -F_a v > 0$$

actuator reaction

Controller

$$L_b i_b^2/2 + C_d V_d^2/2$$

external charger off

Winding and mechanical heat

$$Ri^2, cv^2$$

Coil, converter and logic heat

$$R_b i_b^2, (1 - \eta)P_r, P_0$$